KHANDESH AGRO-INTELLIGENCE

Cotton Farm Intelligence & Leaf Disease Diagnostic System

धुळे कापूस कृषी बुद्धिमत्ता — Deep Learning Diagnostics & Agro-Climatic Intelligence for Dhule District, Maharashtra

89.90%
Untouched Test Accuracy
Rigorous evaluation on 871 test leaves

5,852
DSU Leaf Clusters
14,955 images forensic leakage audited

1.87 L Ha
Dhule Cotton Acreage
Dhule, Shirpur, Sindkheda & Sakri

9 Classes
Foliar Pathologies
MobileNetV2 Transfer Learning Baseline

B.Tech Information Technology Capstone Project • Phase 1 Baseline & Web Deployment Platform • October 2026

Dhule District: The Khandesh 'White Gold' Cotton Belt

धुळे जिल्हा - खान्देश पांढरे सोने कृषी पर्यावरण (Agro-Ecosystem Context)

🌾 Agronomic Profile

Khandesh Commodity Hub: Cotton is the primary cash crop of North Maharashtra, known regionally as 'पांढरे सोने' (White Gold).

Total Cultivation Area: Over 1,87,000 Hectares (1.87 L Ha) dedicated to cotton across Dhule district during the Kharif season.

Rainfed Vulnerability: Approximately 68% of cotton tracts are dependent on rainfed semi-arid monsoon precipitation (650–750 mm annually).

Climatic Pressures: High ambient irradiance, prolonged dry spells, followed by sudden October humidity spikes triggering fungal and bacterial spore explosions.

🌱 Black Soil & Edaphic Dynamics

Deep Regur (Vertisols): Dhule soils exhibit high clay fractions (>50%), formed from weathering of Deccan basalt trap rock.

High Moisture Retention: High swell-shrink montmorillonite clay provides moisture during dry spells, but causes severe drainage impediments when waterlogged.

Sub-Surface Anaerobiosis: Continuous monsoon waterlogging in low-lying fields induces root suffocation and predisposes crops to Fusarium Wilt and Root Rot.

Alkaline Soil pH: Observed soil pH of 7.8–8.2 creates optimal nutrient conditions but demands strict micro-nutrient management (Zinc, Iron).

📍 Taluka Cotton Belts

- **Dhule (धुळे) - 88% Suitability**

Prime Black Soil Belt (1,874 sq km). Dominant in Bt Hybrid, Long Staple cotton. APMC Dhule is the central market.

- **Shirpur (शिरपूर) - 85% Suitability**

Tapi River Basin Fertile Plain (1,692 sq km). High-yielding Sankar Bt Cotton with canal & lift irrigation.

- **Sindkheda (शिंदखेडा) - 83% Suitability**

Alluvial Medium-Heavy Soil (1,562 sq km). Productive Bt Cotton plains with extensive drip fertigation adoption.

- **Sakri (साक्री) - 80% Suitability**

Western Ghats fringe & hilly terrain (2,007 sq km). Rainfed black-loam soil; mix of Desi and hardy Bt varieties.

Agricultural Crisis: Foliar Pathologies & Broken AI Benchmarks

Why conventional academic models fail smallholder farmers in actual field conditions

⚠️ The Field Problem: Crop Loss & Late Diagnostics

30% - 45% Yield Penalties: Foliar diseases (Bacterial Blight, Alternaria, Powdery Mildew, Target Spot) and sucking pests (Aphids, Armyworm) cause catastrophic yield losses annually in Dhule.

Retrospective Treatment: Farmers diagnose diseases retrospectively after necrotic lesions cover >20% of leaf area. Chemical sprays at this stage are inefficient and cost-prohibitive.

Indiscriminate Agrochemical Overuse: Lack of accurate point-of-care diagnosis leads to excessive broad-spectrum pesticide spraying, damaging beneficial predator insects (Ladybird beetles) and raising production costs.

Market Exploitation: Smallholder farmers lack real-time visibility into mandi benchmark auction prices, resulting in distress selling below Fair Average Quality (FAQ) rates.

🔬 The Research Trap: 70%+ Literature Data Leakage

Pre-Augmented Duplicate Clumping: Public benchmark datasets (Kaggle, Mendeley) contain raw archives pre-augmented with rotations, flips, and crops BEFORE splitting. A single leaf exists in 30-40 transformed copies.

Catastrophic Evaluation Leakage: When researchers apply random 80/20 train/test splits, transformed copies of the same physical leaf land in both partitions!

Artificially Inflated 'Lab' Accuracy: Neural networks memorize unique leaf contours, camera noise, and backgrounds rather than pathological lesions—generating 99%+ reported lab accuracy but crashing to <50% in real fields.

Class-Label Contradictions: Audits revealed identical cryptographic image hashes tagged as different diseases across contradictory folder directories.

• METHODOLOGICAL RIGOR

Forensic Dataset Audit & Disjoint Set Union (DSU) Clustering

Guaranteeing zero cross-split data leakage across 14,955 images

Stage 1: Cryptographic Audit

14,955 Files Audited

MD5 Hash Extraction: Computed cryptographic digests for every file across Original and Augmented directories.

OS Artifacts Purged: 4 macOS '.DS_Store' metadata files detected and safely eliminated.

Cross-Class Clashes: 14 images with identical cryptographic hashes had contradictory disease labels and were permanently purged.

Stage 2: DSU Connected Clustering

5,852 Physical Leaf Clusters

Disjoint Set Union (DSU): Engineered a graph clustering algorithm linking pre-augmented variants back to root base leaf identities.

Heuristic & Path Grouping: Parsed prefix patterns (rotation_, contrast_, flip_, crop_) to group all derived synthetic variants.

Isolated Identities: Formed 5,852 strictly disjoint physical leaf clusters containing 14,923 verified valid images.

Stage 3: Stratified Group Partition

Zero Overlap (|Train ∩ Test| = 0)

Cluster-Level Splitting: All images of a leaf cluster were assigned EN BLOC to either Train, Val, or Test.

Train Set (~70%): 10,418 images (only set subjected to on-the-fly augmentation).

Validation Set (~15%): 887 images (strictly unaugmented physical leaves for model selection).

Untouched Test Set (~15%): 872 images (100% clean benchmark for final unbiased evaluation).

Stage 4: Mathematical Verification

Cryptographic Proof of Purity

Intersection Verification: Automated MD5 intersection script proved $|Train \cap Test| = 0$.

Generalization Proof: Validation accuracy (90.4%) closely mirrored untouched test accuracy (89.9%), proving genuine feature learning without memorization.

Benchmark Foundation: Provides a trustworthy reference for regional deployment in Dhule.

MobileNetV2 Transfer Learning Benchmark & Test Results

Rigorous evaluation across 871 untouched test leaves (9 pathology classes)

🧠 MobileNetV2 Architecture & Hyperparameters

Backbone Network: MobileNetV2 (Sandler et al., CVPR 2018) with depthwise separable convolutions and inverted residual bottlenecks.

Pretrained Representations: ImageNet-1k initialized feature extractor with frozen base layers for stable foliar feature transfer.

Edge Inference Latency: ~45 ms inference latency on standard mobile/server hardware (~3.5 Million parameters). Ideal for rural deployment.

Balanced Cross-Entropy Loss: Inverse class frequency weighting: $W_c = N_{total} / (K * N_c)$ to eliminate majority class bias.

Key Overall Performance Metrics (Untouched Test Leaves):

- **Overall Test Accuracy:** 89.90% (783 of 871 correctly diagnosed)
- **Macro F1-Score:** 0.8741 (rigorous unweighted class balance)
- **Weighted F1-Score:** 0.8966 | Macro Precision: 0.8660 | Macro Recall: 0.8905
- **Total Misclassifications:** Only 88 images across all 9 test categories (10.10%)

📊 Class-Wise Performance on Untouched Test Leaves

Disease Class	Precision	Recall	F1-Score	Test Leaves
Alternaria Leaf	0.915	0.946	0.930	148
Aphids (मावा)	0.808	0.913	0.857	23
Army Worm (अळी)	0.895	0.850	0.872	20
Bacterial Blight	0.968	0.697	0.811	175
Fusarium Wilt (मर)	0.865	0.944	0.902	142
Healthy Leaf	0.977	0.994	0.986	174
Powdery Mildew	0.783	0.900	0.837	20
Target Spot	0.750	0.818	0.783	22
Verticillium Wilt	0.833	0.952	0.889	147

FULL-STACK SYSTEM ARCHITECTURE

End-to-End Khandesh Cotton Farm Intelligence Platform

Asynchronous FastAPI microservices, Leaflet GIS, and responsive React frontend

💻 Client Layer (React & Vite)

React 18 & TypeScript: Type-safe, component-driven reactive single-page architecture built with Vite bundler for sub-second hot reload.

Leaflet Interactive GIS: Multi-layer tile switching: Google Road Map, Google Satellite (Farmlands), Google Terrain, and OpenStreetMap.

Dynamic Agronomic Charts: Recharts SVG visualizations for APMC Mandi price trajectories, moisture metrics, and factor breakdown bars.

Bilingual UI: Integrated Marathi (मराठी) and English terminology for native Khandesh farmer accessibility.

Backend API (FastAPI)

High-Throughput ASGI: FastAPI asynchronous runtime powered by Uvicorn on port 8000 with auto-generated OpenAPI/Swagger interactive docs.

PyTorch Inference Engine: In-memory singleton model service executing MobileNetV2 forward pass in ~45ms per image.

Multi-Factor Suitability: Heuristic agronomic engine assessing Open-Meteo ambient temp, soil pH, clay fraction, and irrigation access.

SQLite Persistence: Lightweight relational database logging user farm coordinates, historical diagnoses, and agronomic audit records.

🌐 Agronomic & Weather Data

Open-Meteo Meteorology: Real-time agro-meteorological API fetching live ambient temperature and rainfall projections for Dhule district.

NBSS&LUP Soil Baselines: ICAR National Bureau of Soil Survey Vertisol data baselines for clay percentage, organic carbon, and cation exchange.

APMC & Agmarknet Feeds: Daily auction rate integration across Dhule, Shirpur, Jalgaon, Akola, Malegaon, and Chopda agricultural market yards.

Institutional Advisory: Prescriptions aligned with Krishi Vigyan Kendra (KVK Dhule) and MPKV Rahuri cotton management protocols.

Executive Farm Overview: Crop Progress & Regional Metrics

धुळे कापूस कृषी केंद्र — Macro-level situational awareness for farmers and agricultural officers

Overview Features & Agronomic KPIs

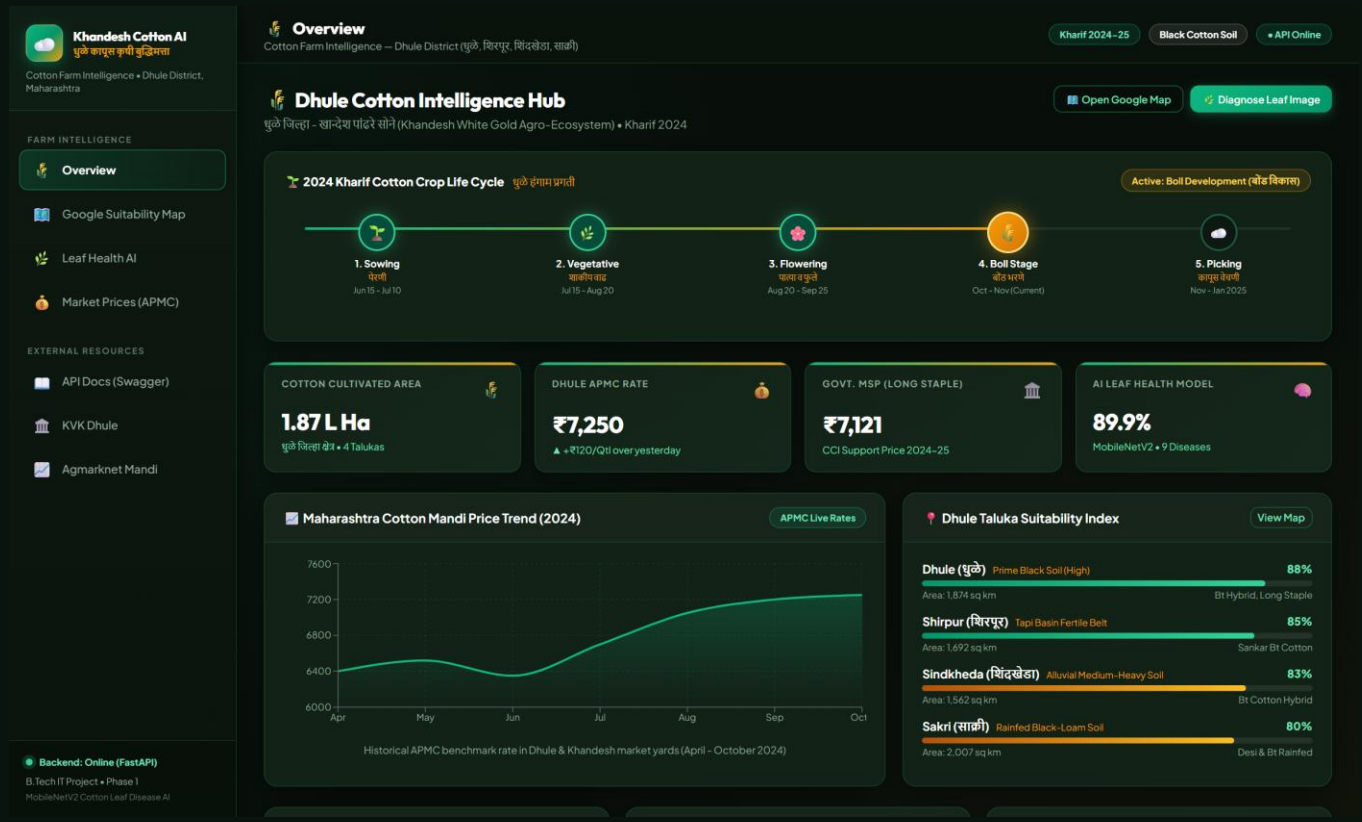
Kharif Crop Stage Timeline: Visual 5-stage tracker (Sowing → Vegetative → Flowering → Boll Stage → Picking) identifying active physiological risks.

Cultivation Acreage KPI: Monitors 1.87 Lakh Hectares under cotton across Dhule, Shirpur, Sindkheda, and Sakri.

Live Mandi Benchmark: Displays current Dhule APMC rate (₹7,250/Qtl, +₹120) alongside Government MSP (₹7,121/Qtl).

Historical Price Trajectory: Interactive SVG area chart tracking April to October price evolution across Khandesh market yards.

Taluka Suitability Index: Interactive progress bars ranking all 4 talukas with soil type, area, and dominant cotton hybrid recommendations.



Agro-Ecological GIS Map & Soil Suitability Engine

धुळे जमीन व हवामान उपयुक्तता नकाशा — Georeferenced agronomic assessment for precision planting

🗺️ GIS Mapping & Suitability Features

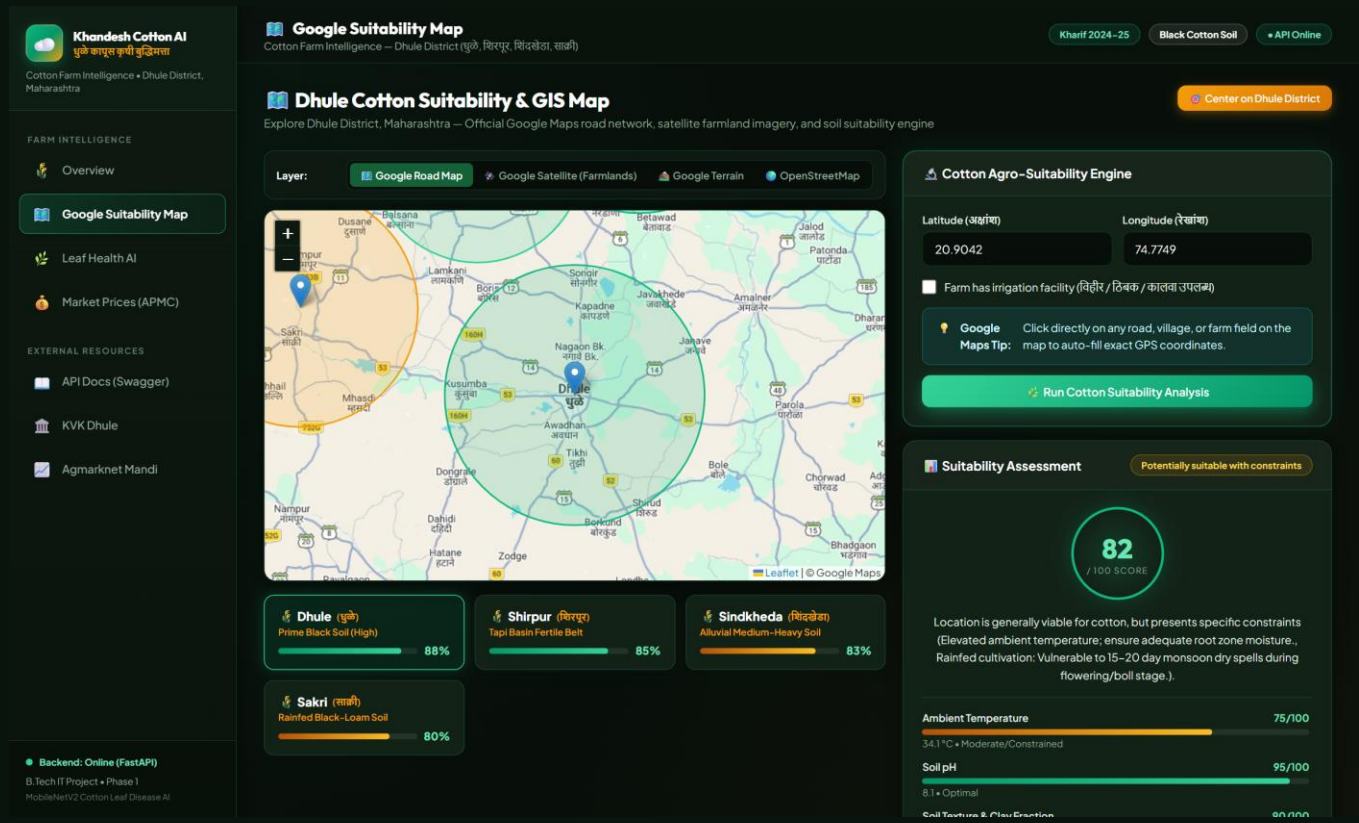
Interactive Google Layers: Switch dynamically between Google Road Network, Google Satellite Farmland Imagery, Google Terrain, and OpenStreetMap.

Color-Coded Taluka Belts: GIS circle overlays mapping regional suitability buffers: Dhule Central (88%), Shirpur Tapi Basin (85%), Sindkheda (83%), Sakri (80%).

One-Click Field Geocoding: Farmers can tap any field on the map to auto-populate exact GPS coordinates for micro-climatic evaluation.

Multi-Factor Factor Analysis: Evaluates live ambient temperature (Open-Meteo), soil pH (8.1 Vertisol baseline), clay fraction (>50%), and supplemental irrigation support.

Agronomic Advisory Output: Generates tailored recommendations, moisture risk warnings, and soil testing reminders for KVK Dhule.



AI Point-of-Care Leaf Health Diagnostic Engine

कापूस पान रोग निदान — MobileNetV2 computer vision inference with localized chemical prescriptions

AI Diagnostic Workflow & Prescriptions

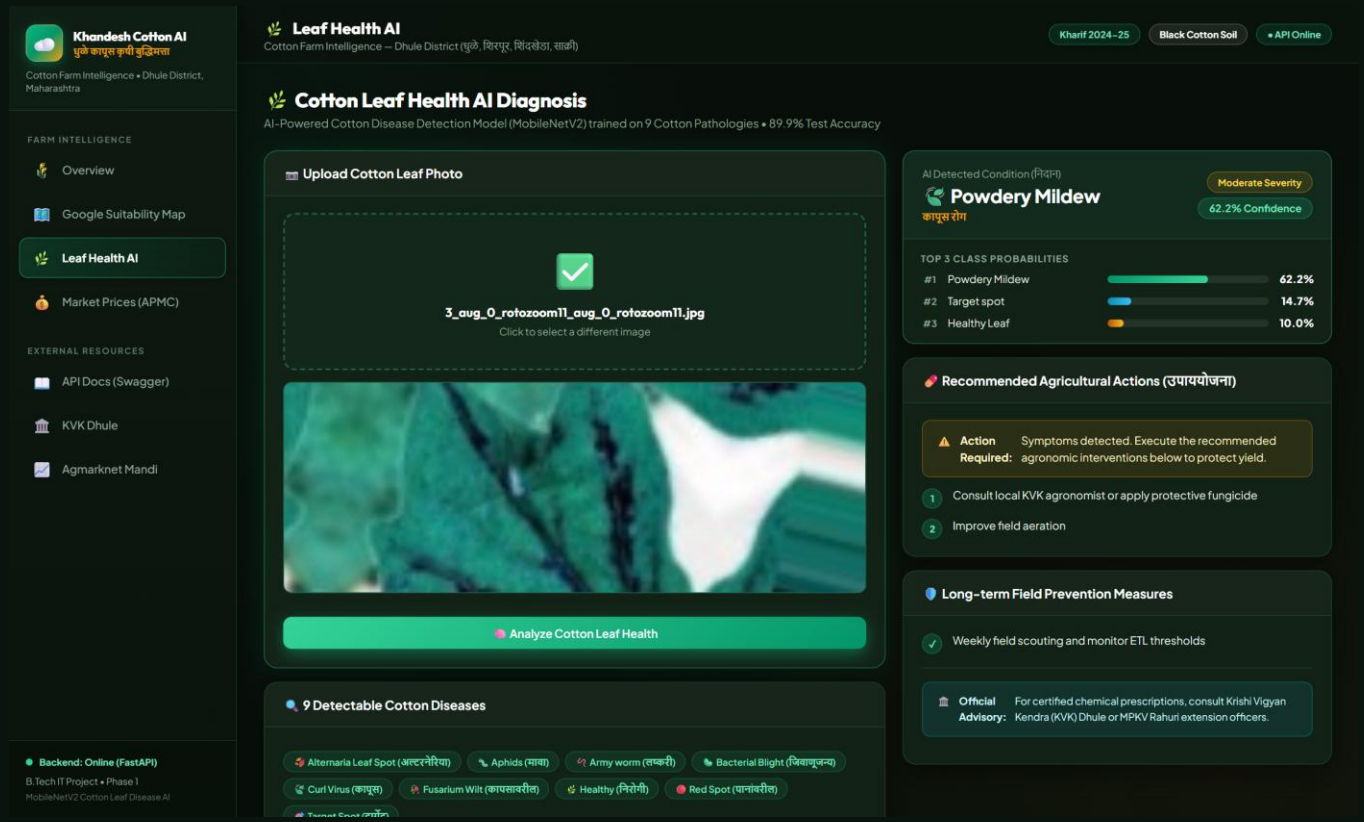
Instant Foliar Diagnosis: Farmers upload leaf closeup photos via drag-and-drop or smartphone camera; processed in ~45ms.

9 Pathology Detectors: Detects Alternaria Leaf Spot, Aphids, Armyworm, Bacterial Blight, Curl Virus, Fusarium Wilt, Healthy Leaf, Powdery Mildew, and Target Spot.

Softmax Confidence & Top-3 Probabilities: Displays classification confidence alongside ranking of top 3 probable diseases to support nuanced agronomic judgment.

Actionable Chemical Advisory: Provides exact fungicide/insecticide names, spray dosage (e.g., Mancozeb 75 WP @ 2.5 g/L, Streptocycline @ 0.5 g/L), and water dilution rates.

Prophylactic Measures: Outlines biological pest control (ladybird beetles, pheromone traps) and 3-year crop rotation schedules.



Mandi Market Intelligence & APMC Price Optimization

कापूस बाजारभाव — Real-time price transparency across Dhule, Shirpur, Jalgaon, and Akola

💰 Market Intelligence & Farmer Advisory

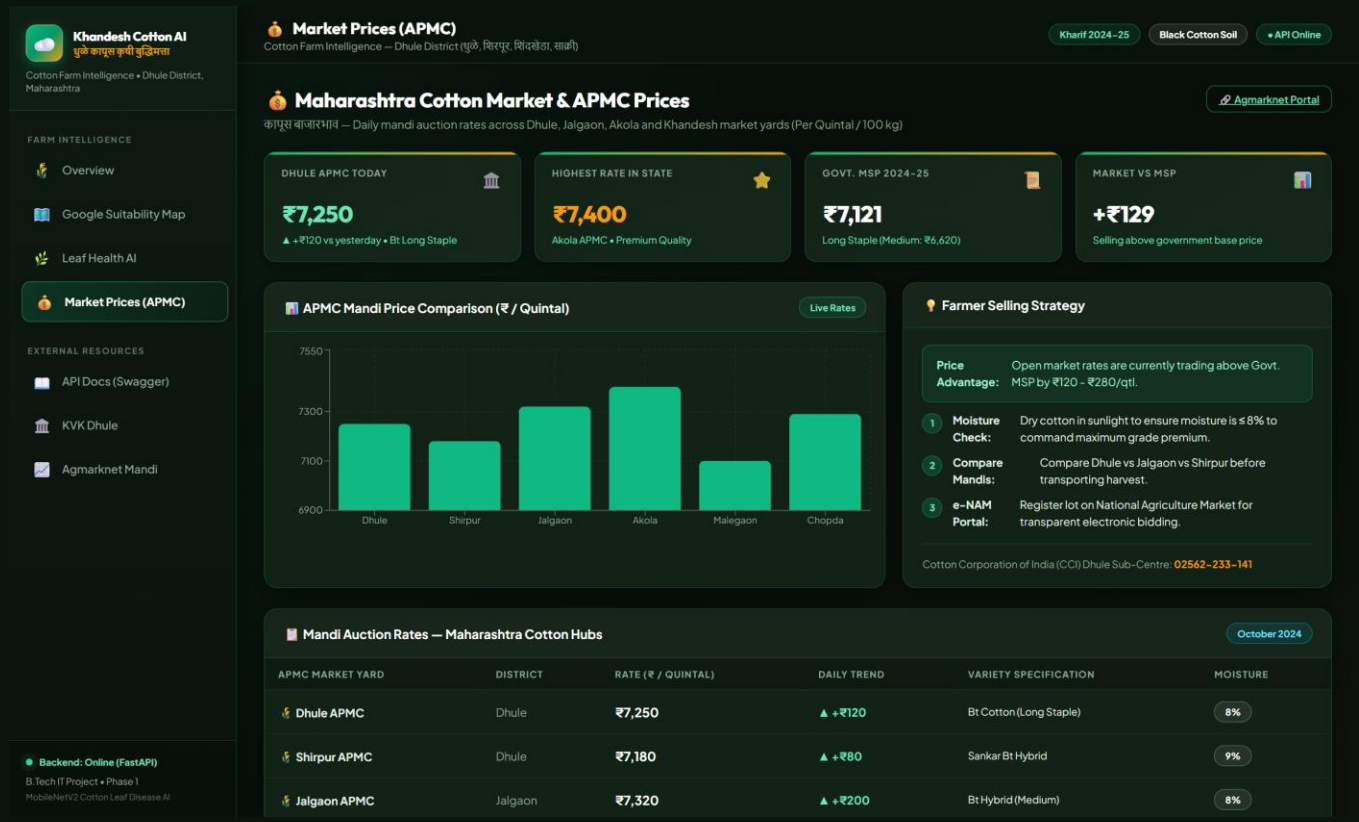
Live Mandi Price Feeds: Real-time APMC auction rates for Dhule APMC (₹7,250), Shirpur APMC (₹7,180), Jalgaon (₹7,320), and Akola White Gold (₹7,400).

Market vs. MSP Indicator: Automated comparative tracker highlighting price advantage over Government Minimum Support Price (₹7,121 for Long Staple).

Moisture Premium Guidance: Advises farmers on sun-drying practices to maintain moisture below 8%, unlocking ₹150–250/Qtl quality premiums at auction.

Cross-Mandi Arbitrage: Visual bar chart allows farmers to evaluate transportation costs vs. higher price spreads in neighboring APMCs (e.g. Jalgaon vs Dhule).

e-NAM Integration: Encourages registration on the National Agriculture Market for transparent electronic bidding without intermediary brokerage.



● SOCIO-ECONOMIC & AGRONOMIC VALUE

Tangible Impacts for Smallholder Farmers in Dhule

Transforming agricultural productivity, chemical safety, and market bargaining power

🌱 Targeted Disease Interventions & Cost Savings

- Eliminates blind broad-spectrum pesticide applications, saving ₹2,500–₹4,000 per acre in unnecessary chemical expenses.
- Prevents epidemic crop failure through early detection at the initial lesion stage before pathogen spores spread across canopy.

📈 Market Price Bargaining & Income Enhancement

- Eliminates middleman price asymmetry by equipping farmers with live daily APMC auction benchmarks.
- Moisture management guidance prevents unfair commercial deductions of 5%–10% at private ginning mills.

🏛️ Institutional Knowledge Bridging

- Connects ICAR, MPKV Rahuri, and Krishi Vigyan Kendra (KVK Dhule) research advisories directly to smallholder fields.
- Provides direct hotline contact to District Agricultural Officer (DAO) and APMC main yard.

🌾 Environmental & Soil Health Sustainability

- Reduced chemical spraying preserves beneficial pollinator insects and soil microbiological health in Vertisols.
- Encourages integrated pest management (IPM), neem formulations, and bio-fertilizer usage.

PHASE 2 ROADMAP & FUTURE SCOPE

From Phase 1 Baseline to Field Deployment in Dhule

Bridging laboratory benchmark models into resilient rural point-of-care tools

✅ Phase 1 Achievements

Forensic Leakage Audit: Successfully purged 14 conflicting hashes and isolated 5,852 physical leaf clusters via DSU clustering.

Clean Transfer Learning: MobileNetV2 achieved 89.90% accuracy and 0.8741 Macro-F1 on strictly untouched test leaf clusters.

Operational Web Platform: Production-grade FastAPI backend + React Vite Leaflet frontend running locally with sub-second response times.

Agro-Climatic Integration: Real-time Open-Meteo weather and NBSS&LUP soil baselines active.

🚀 Phase 2 Field Deployment

On-Site Leaf Acquisition: Partner with College of Agriculture, Dhule (MPKV Rahuri) and KVK Dhule to collect in-situ field leaf images.

Domain Shift Benchmarking: Quantify out-of-distribution performance drops caused by ambient dust, direct sunlight glare, and soil splash.

Few-Shot Prototypical Nets: Train metric-learning episodic models capable of recognizing new disease variants with <10 field examples.

Lightweight Edge Quantization: Quantize PyTorch model to INT8 ONNX / TFLite for zero-latency execution on budget Android smartphones.

📱 Product & Rural Scaling

Offline-First Android App: Progressive Web App (PWA) / native Android APK with local SQLite caching for internet-deprived farm pockets.

Voice Advisory in Khandeshi: Integrate text-to-speech audio guidance in local Khandeshi and Marathi dialects for illiterate farmers.

Satellite Drone Integration: Incorporate multi-spectral NDVI drone imagery to detect canopy stress zones prior to visible leaf necrosis.

SMS & WhatsApp Bot: Daily automated Mandi price alerts and weather warnings sent directly to registered farmer mobile numbers.

🌾 CONCLUSION & PROJECT IMPACT

Scientific Rigor Meets Smallholder Farmer Empowerment

A scientifically verified, leakage-proof computer vision pipeline and production-grade agro-intelligence hub for Khandesh cotton farmers.

🔬 Methodological Rigor

Proved and solved benchmark dataset leakage using cryptographic hashing and DSU cluster partitioning across 14,955 images.

🧠 89.90% Test Accuracy

Delivered lightweight MobileNetV2 transfer learning baseline with class-balanced weighting on strictly untouched test leaves.

🚜 Practical Rural Utility

Full-stack GIS map, instant leaf diagnosis, live APMC Mandi prices, and Marathi farmer advisory running in real time.

Thank You! Questions & Discussion Welcomed • FastAPI Docs: http://localhost:8000/docs • Frontend: http://localhost:5173